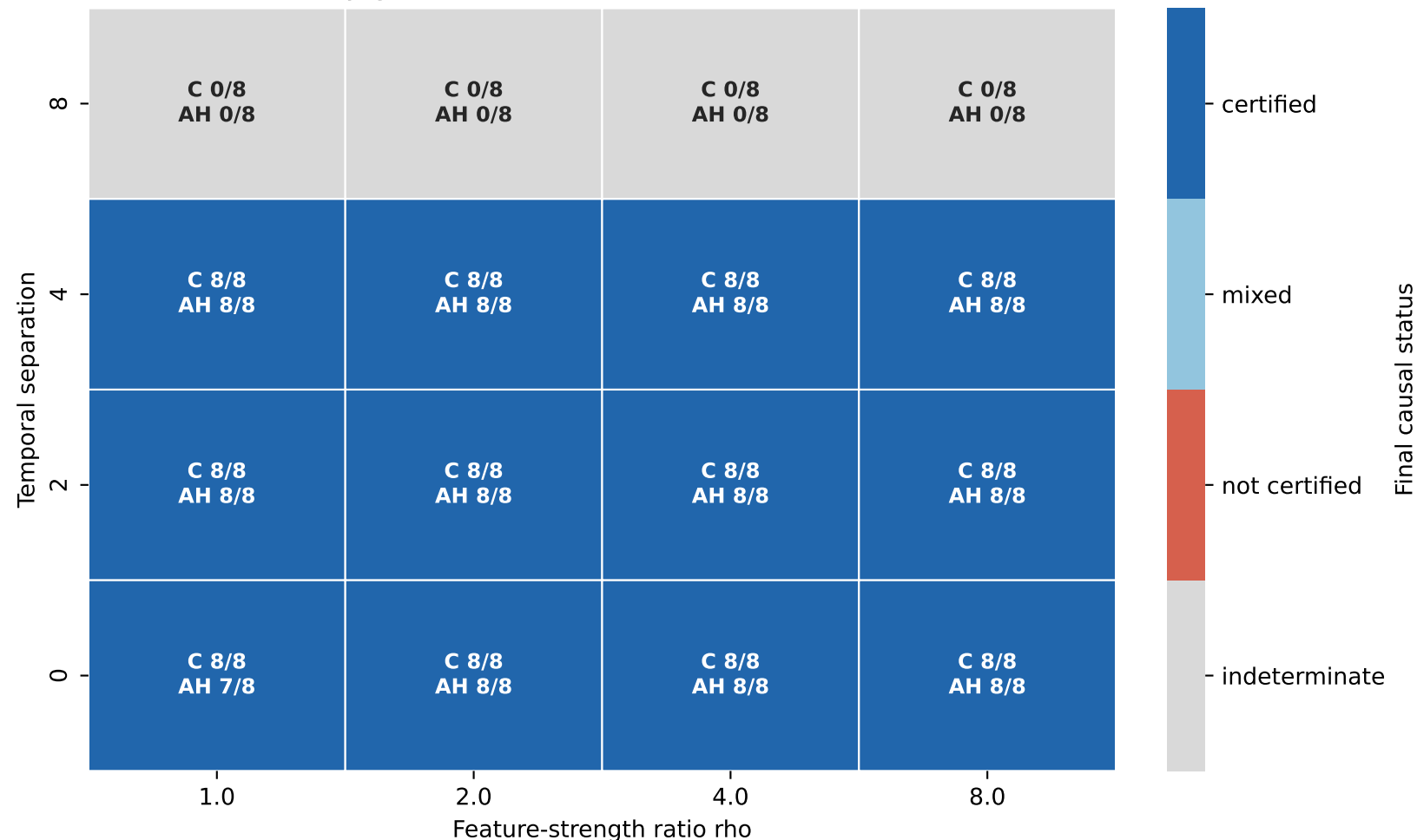


Learnability-gated causal starvation and at-hit robustness



C = causal certificate (negative gap at some positive time); AH = stricter negative gap at the weak-only first hit.